

Design and Evaluation of a 650 V, 300 A GaN-Based Power Module with Integrated Drivers and Ultra-Low Inductance Layout

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Abstract— Gallium Nitride (GaN) devices offer high switching speed, low loss, and compact design. To fully exploit these advantages, packaging is critical. In this paper, a 650 V/300 A half bridge GaN power module is designed and packaged. The module features a multilayer substrate design, embedded decoupling capacitors, and co-packaged gate drivers, enabling a fast switching speed with reduced parasitic inductances. A low power loop inductance with about 40% reduction (~5.52 nH) in the proposed design is achieved compared conventional single-layer substrate design (9.1 nH). Thermal performance is analyzed with finite element analysis (FEA) to evaluate the high current operation. Experimental results validate the module concept with fast switching and reduced switching losses. The proposed design demonstrates a path toward reliable high-current GaN modules for electric vehicle chargers and industrial systems.

Keywords— Gallium Nitride (GaN), multilayer substrate, Q3D, high current operation, finite element.

I. INTRODUCTION

Gallium Nitride (GaN) HEMT technology has seen rapid progress in recent years. Manufacturing costs have steadily decreased, making these devices increasingly accessible. Today, 650 V GaN devices are becoming strong alternatives to conventional silicon MOSFETs in low- and medium-voltage power conversion applications. Compared to silicon, GaN devices offer higher switching speeds, lower switching losses, and greater power density. These characteristics allow for more compact and efficient designs. As a result, GaN devices are being widely adopted in data centers, on-board chargers (OBCs), renewable energy systems, and motor drive applications. Their fast switching capability also enables improved system dynamics and reduced cooling requirements, supporting the trend toward more energy-efficient power electronics.

To fully utilize the advantages of GaN devices, the packaging of high-current GaN power modules has become a critical research area in both academia and industry [5]. Leading manufacturers, including Infineon, GaN Systems, Transphorm, and EPC, are actively developing advanced module platforms. These designs often combine multiple GaN dies in parallel to achieve current ratings that exceed 150 A [6]–[8]. Such parallelization is essential for meeting the demands of high-power applications while maintaining compact form factors.

Academic groups have also introduced innovative approaches. Several research efforts have focused on reducing parasitics through the use of embedded substrates, planar interconnection schemes, and integrated decoupling

capacitors. These techniques enable low-inductance half-bridge configurations, which are crucial for GaN devices operating at high switching speeds. The reported modules demonstrate fast and stable switching with minimized voltage overshoot and reduced ringing [9]–[11]. Beyond performance, these designs also highlight the importance of packaging in ensuring system reliability, thermal management, and electromagnetic compatibility.

Integration has become an important trend in GaN module development. Power modules such as Infineon's CoolGaN™ devices and Navitas's integrated GaN platforms, combine multiple functions into a single package. These include gate drivers, protection features, and monitoring circuits. By bringing these elements together, these modules can achieve faster switching performance, simplified system design, and improved reliability.

Despite these advantages, current efforts have been mostly limited to low and medium power applications. Only a few studies have focused on high-current GaN modules that can operate under the demanding requirements of electric vehicle on-board chargers (OBCs), DC fast chargers, and industrial power conversion systems [12–20]. Developing such modules remains a challenge due to the combined needs of thermal management, low-inductance packaging, and system-level integration. Addressing these challenges will be critical for enabling GaN technology to expand into high-power, high-current domains.

This work presents a highly integrated 650 V GaN power module rated at 300 A. The design features ultra-low parasitic inductance and supports high switching speeds. A multi-layer substrate, embedded decoupling capacitors, and co-packaged gate drivers are heterogeneously integrated. This approach minimizes both power and gate loop inductances, enabling reliable fast switching and significantly improving efficiency and power density at the system level.

II. MODULE DESIGN

To meet the design goals of high integration and faster switching, a multilayer substrate GaN power module is proposed. The exploded view of the module is illustrated in Fig. 1(b). The internal structure of the module is shown in Fig. 1(b). The module has compact dimensions of 80 mm × 65 mm × 25 mm. Both DC+ and DC− terminals are located on the same side, which simplifies system integration and reduces layout complexity. The module is designed as a half-bridge configuration, featuring integrated decoupling capacitors and built-in gate drivers. Each switch position incorporates two 650 V GaN HEMTs connected in parallel,

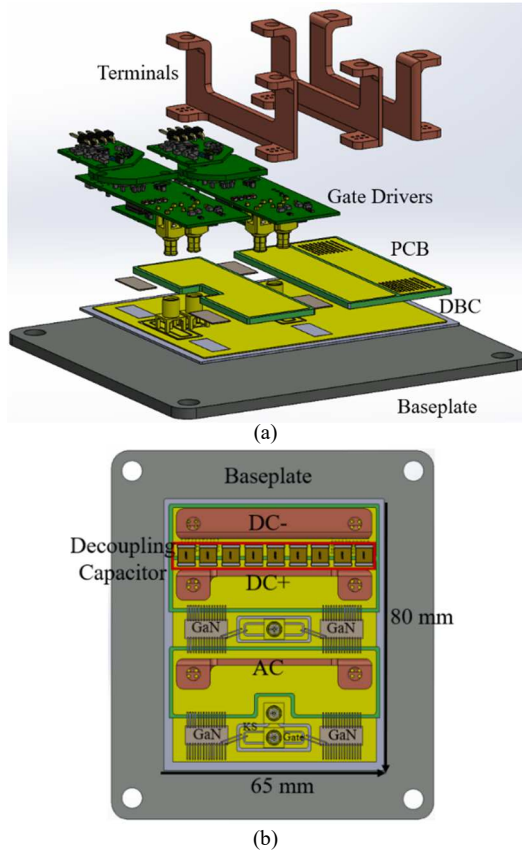


Fig. 1: Module design concept (a) Exploded view of the GaN module (b) Module size and internal structure.

Table I: Module Specification

Design	Value
Module dimensions	80 mm × 65 mm × 25 mm
Topology	Half bridge
Chip type	Infineon GS-065-150V-7-D6
Switch configuration	2 chips per switch position
Module integration	- Multilayer Substrate - Embedded decoupling capacitor - Built in gate driver

enhancing current handling and reducing conduction losses. The proposed 650 V, 300 A module employs MCX series coaxial connectors, which provide robust mechanical support for the integrated gate drivers and ensure reliable signal connections. This design allows for high power density while maintaining thermal efficiency and low parasitic inductance. The module is well-suited for high-performance applications such as on-board chargers, data center power supplies, and industrial module converter. The main features and specifications of the proposed multilayer substrate 650 V GaN power module are summarized in Table I.

III. MODULE FABRICATION

The fabrication of the 300 A GaN power module was carried out in seven main steps. First, the GaN dies and PCB components were soldered using SMD291SNL10 solder paste, with reflow conducted at 249 °C to ensure reliable attachment. Following die assembly, wire bonding was performed with a Hesse BJ935 heavy aluminum wire bonder. Twelve-mil aluminum wires were used for the source connections, while finer wires were used for the gate terminals to minimize parasitic inductance. Power terminals and coaxial

Table II: Fabrication Process of the GaN Module

Step	Processes to be used in each Fabrication Steps	3D model
FS 1	- Soldering of PCB and GaN chip using Sikama. - SMD291SNL10 solder paste will be used for attachment. - Reflow temperature 249 °C.	
FS 2	- Wire bonding of the module - Hesse BJ935 Heavy Aluminum Wire Bonder to be used for wirebonding. 12 mil wire bonds to be used.	
FS 3	- Soldering of Coaxial connectors and Power terminals. - TS391LT10 solder paste will be used for attachment. - Reflow temperature 165°C	
FS 4	- Soldering of baseplate with the module - TIM will be used for attachment	
FS 5	- Attach the side part of the case on the DBC - RTV167 adhesive is used for attachment. - Curing time 24 hrs at room temperature.	
FS 6	- Wacker SilGel 612 A/B encapsulant to be used. - Exact dispensing vacuum to be used. - Filling and degassing twice before curing. - Attach the top part of the case. - RTV167 adhesive is used for attachment.	

connectors were then mounted onto the PCB using TS391LT10 solder paste, with reflow soldering carried out at 165 °C.

Next, a thermal interface material (TIM) was applied to secure the module onto the baseplate, providing effective thermal conduction. The sidewalls of the case were attached to the DBC substrate using RTV167 adhesive, followed by a curing period of 24 hours at room temperature to ensure mechanical stability. Encapsulation was performed using Wacker SilGel 612 A/B. A vacuum dispensing method was applied, with the filling and degassing steps repeated twice to eliminate trapped air and ensure uniform coverage. Finally, the top cover of the case was mounted using RTV167 adhesive, completing the module assembly. To validate the proposed module concept, a PCB based module is developed with 2 oz copper layers. The fabrication process of the module is shown in Table II.

IV. THERMAL ANALYSIS VIA FEA MODEL

A finite element analysis (FEA) model was created to evaluate the thermal performance of the high-current GaN module. The model used the material properties summarized in Table III. A heat transfer coefficient of 1000 W/m²·K was applied to represent forced liquid cooling and estimate the junction temperature of the devices.

Each GaN die in the module operates at a current level of

Table III: Thermal Parameters of 300 A GaN Module

Parts	Thermal Conductivity (W/m·K)
GaN	230
Silver Sinter	280
Copper	398
AlN	310

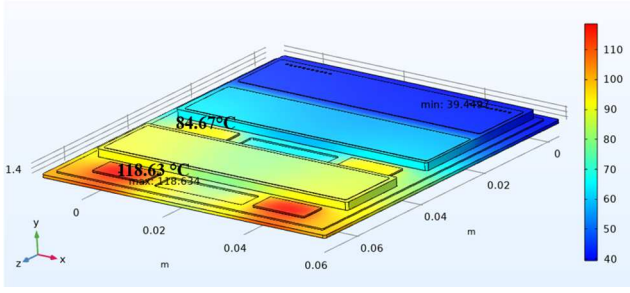


Fig. 2: Thermal Analysis of the module with each die having power loss of 225 W.

150 A. At this rating, the devices generate substantial heat due to conduction losses. For a die with an on-resistance (R_{dson}) of 10 m Ω , the estimated power loss is approximately 225 W per device. This loss translates into a significant rise in junction temperature, as illustrated in Fig. 2. Elevated junction temperature can severely impact device reliability, switching performance, and overall system efficiency.

To mitigate this issue, robust thermal management is essential. Conventional air cooling or basic heat sinks are not sufficient for such high power densities. Advanced cooling strategies, such as liquid cooling or two-phase heat transfer systems, may be required to maintain safe operating conditions. Optimized thermal interface materials and package architectures can also enhance heat spreading and reduce thermal resistance. Without effective thermal management, the module is at risk of overheating, which can shorten device lifetime and compromise system-level performance.

V. PARASITIC INDUCTANCE SIMULATION

Minimizing parasitic inductance is one of the primary goals in power module packaging. For GaN devices, reduced inductance directly improves switching performance. It also lowers voltage overshoot, decreases device stress, and helps suppress electromagnetic interference (EMI).

To evaluate the proposed design, a Q3D finite element analysis was carried out using ANSYS. Power loop was extracted. The analysis was performed at a reference frequency of 10 MHz, which is commonly used for high-frequency module evaluation. The results show that the power loop parasitic inductance is 5.52 nH. This confirms that the proposed architecture provides a significant reduction in loop inductance. This is due to the magnetic flux cancellation, as the current loop flows in opposite direction. The current loop of the designed module is shown in Fig. 3.

For comparison, a conventional single-layer design exhibits a power loop inductance of 9.1 nH, as summarized in Table IV. The reduction achieved demonstrates the effectiveness of the multilayer packaging structure. Lower inductance not only enables faster switching but also improves system efficiency and reliability, especially under high current operation.

VI. EXPERIMENTAL VERIFICATION

To verify the proposed module concept, a prototype was first fabricated on a PCB that reproduced the layout of the target multi-layer substrate. The PCB served as a low-cost platform for preliminary evaluation before moving to the final design. One GaN die was mounted on the high-side and another on the low-side, as shown in Fig. 4. This configuration allowed

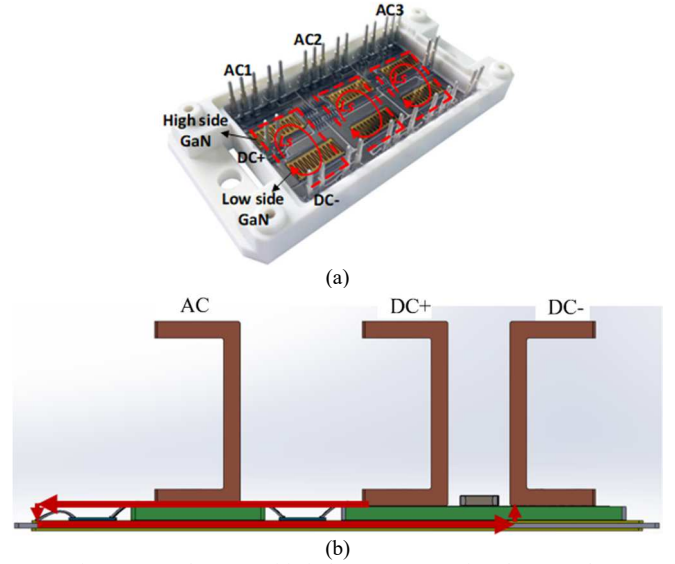


Fig. 3: Power loop parasitic inductance comparison between the conventional GaN power module and proposed GaN power module. (a) Power loop layout of conventional GaN power module ($L_s=9.1\text{nH}$); (b) Power loop layout of proposed GaN power module ($L_s=5.52\text{nH}$)

Table IV: Power Loop Inductance comparison of 300 A GaN Module

	Power Loop inductance
Multilayer architecture with terminals	5.52 nH
Single layer architecture with terminals	9.1 nH

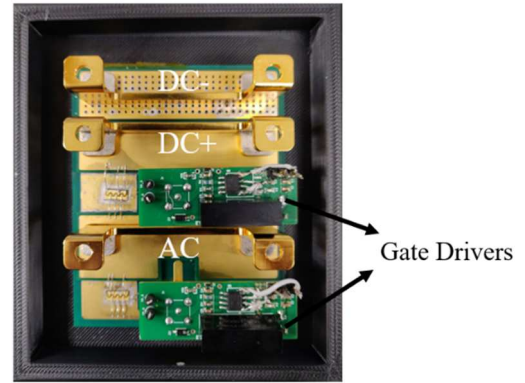


Fig. 4: Picture of the developed GaN Module. validation of the basic half-bridge structure and assessment of parasitics in a simplified form. The prototype also provided an opportunity to test assembly procedures and ensure compatibility with the planned integration approach.

A. Static Characterization

The static characterization of the solder-ball-connected GaN die was performed using a curve tracer. This method enabled the evaluation of key electrical parameters, such as the output characteristics and the on-resistance (R_{dson}). The measured R_{dson} was approximately 20 m Ω , which is consistent with expected device performance. These results confirm the electrical feasibility of the packaging approach and validate proper die-to-substrate integration. Due to probe limitations, the maximum drain current during testing was restricted to 4 A. As a result, the measurements primarily reflect the device behavior under low-current conditions. The results provided insight into device linearity, conduction

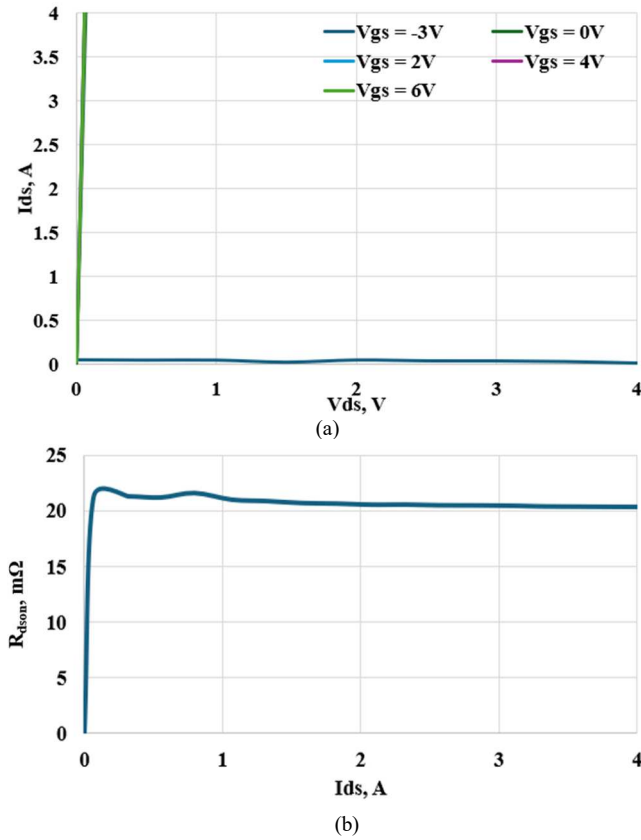


Fig. 5: Static Characterization of the GaN Module, (a) Output Characteristics of the GaN die, (b) On state resistance of the GaN die.

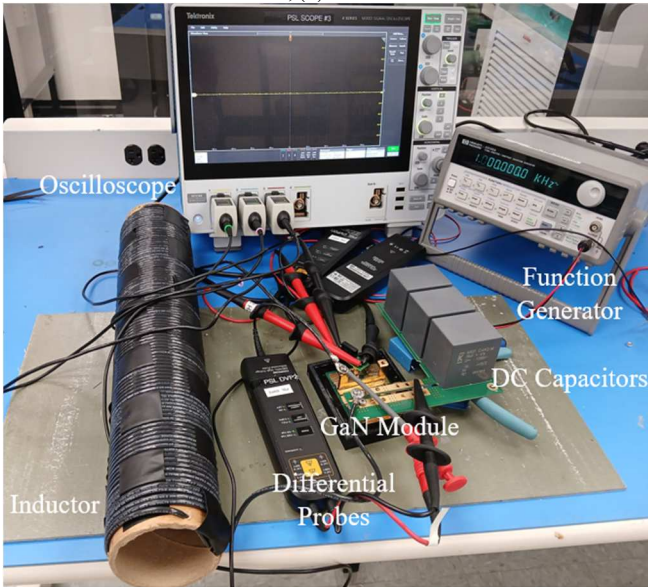


Fig. 6: Double Pulse experiment setup of the GaN Module.

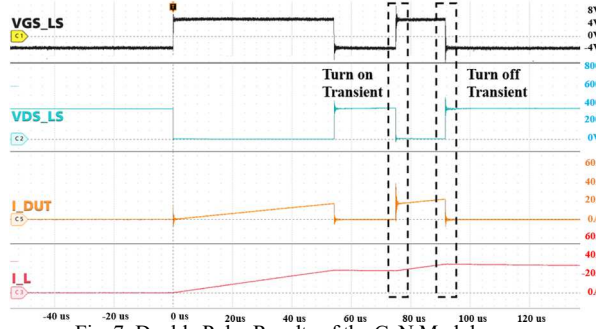


Fig. 7: Double Pulse Results of the GaN Module.

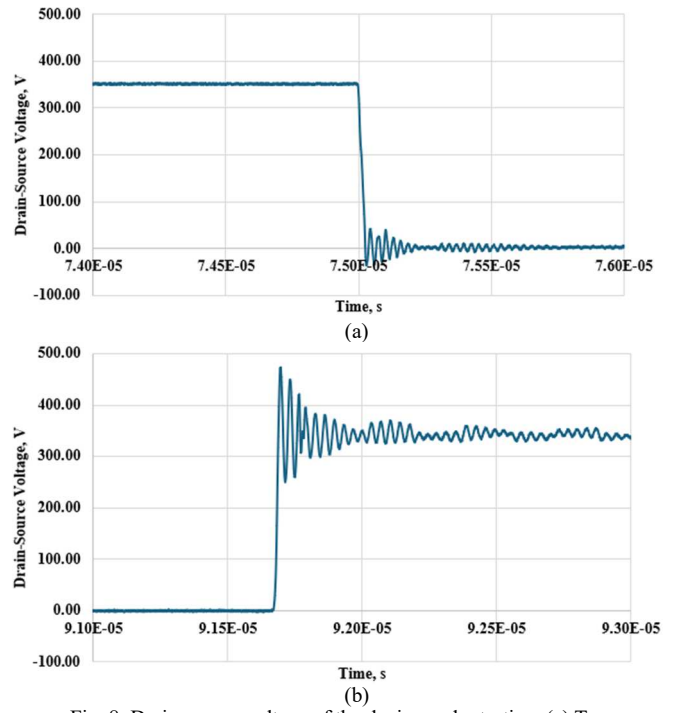


Fig. 8: Drain-source voltage of the device under testing, (a) Turn-on transient, (b) Turn-off transient.

behavior, and packaging reliability at the initial level. The static characteristics results are shown in Fig. 5.

B. Dynamic Characterization

To evaluate the proposed design, a Q3D finite element A double-pulse test (DPT) was carried out on the fabricated GaN module to evaluate its dynamic switching behavior. The experimental setup is illustrated in Fig. 6. In this configuration, the high-side device was held at a constant gate bias of -3 V, allowing it to operate as a freewheeling diode. The low-side device was then driven with two consecutive short pulses to observe its turn-on and turn-off transitions.

The test was performed with a DC bus voltage of 350 V. The corresponding waveforms are presented in Fig. 7. The measured signals include the gate-to-source voltage of the low-side switch, the drain-to-source voltage (V_{DS}) of the low-side switch, and the inductor current. These results provide direct insight into the switching dynamics, voltage overshoot, and current response of the module under realistic operating conditions. The data also confirm the capability of the proposed packaging design to support fast switching transitions while maintaining stable operation.

Figure 7 shows the transient turn-on and turn-off the device under testing (low side GaN switch).

VII. CONCLUSION

A 650 V, 300 A GaN power module has been designed and demonstrated. The module uses a multilayer substrate with embedded capacitors and co-packaged gate drivers. This approach reduces parasitic inductance to 5.52 nH from 9.1 nH which is the power loop inductance of a single layered design. Inductance extraction confirms significant reduction compared to a single-layer design. The simulated maximum junction temperature of the module is 118.63 °C with 225 W losses under a heat transfer coefficient of 1000 W/m²·K. Experimental tests validate stable switching and reliable operation. The results highlight the potential of the proposed

module for EV chargers and industrial applications. It also demonstrates the feasibility of compact, high-current GaN packaging. Further work will focus on reliability evaluation and advanced cooling strategies.

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